📌 eSim Tool Manager (CLI)

eSim Summer Fellowship 2026 – Task 5
Name: Pranaw Somesh R

Department: Computer and Communication Engineering

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1. Project Overview

This project is a Command Line Interface (CLI)-based Tool Manager developed using Python for the eSim Summer Fellowship 2026.

The main objective of this project is to simplify the setup and management of circuit simulation tools such as **ngspice** and **KiCad**. The tool allows users to check whether a tool is installed and automatically install it if required.

The system is designed specifically for Linux environments and interacts directly with the system using command-line utilities.

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2. Features

- Check whether a tool is installed
- Automatically install tools using system package manager
- Display all supported tools
- Check multiple tools at once
- Handle unsupported tools gracefully
- Provide user-friendly CLI menu
- Colored terminal output for better readability
- Logging system for tracking actions

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3. Tech Stack

- Python
- Linux (Ubuntu)
- Command Line Interface (CLI)
- Python modules:
    - os
    - colorama

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4. Project Architecture

The project follows a modular structure for better organization:

- main.py → Handles user input and menu system
- installer.py → Handles installation of tools
- checker.py → Checks tool availability and version
- utils.py → Logging functionality

This modular design makes the system easy to maintain and extend.

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5. Working Flow

The system takes user input through a menu-driven interface and performs operations accordingly.

Flow:

User → CLI Menu → main.py → installer/checker → system commands → output

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6. Commands Used

The project runs through an interactive menu:

python3 main.py

Example operations:

- Check tool installation
- Install tool
- Display supported tools
- Check all tools

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7. Sample Output

Check Tool:

htop is installed ✅

Not Installed:

abcxyz is NOT installed ❌
Do you want to install it? (y/n):

Install Output:

⚙️ Installing htop...
✅ htop installed successfully

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8. Demo

A demo screenshot and video are included in the GitHub repository to demonstrate the working of the system.

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9. Conclusion

This project demonstrates the implementation of a CLI-based automation tool for managing system dependencies. It simplifies the process of installing and verifying tools required for circuit simulation.

The project can be further improved by adding support for more tools, implementing cross-platform compatibility, and developing a graphical user interface.